

NJIT

New Jersey Institute
of Technology

Interpretable Electrode-Band Optimization for Affective Emotion Decoding

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Agenda

- Intro
- Study Overview
- Study
 1. Frequency Domain Transformation for Structured Modeling
 2. Modeling: Superiority of Customized Residual CNN
 3. Explainability with SHAP: Identifying Key Brain Regions and Bands
 4. Insights from Physiological Signals
 5. Cross-Dataset Generalization: SEED-V and AMIGO
- Key Takeaways

Intro: Who Is This Guy??



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- Hurlburt Chair Professor
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Intro: Our Lab at NJIT

Welcome to Our Lab

The Martin Tuchman School of Management is home to our cutting-edge Neuro-Analytics in Business Lab. We engage in research and technology development that helps address issues facing data science, consumer behavior, behavioral finance, and the future of work. Our lab involves cutting-edge interdisciplinary research at the Martin Tuchman School of Business, and does so in a collaborative community environment.



Our Team



Intro: Our Lab at NJIT

Comparison of Techniques to Enhance Human Preference Prediction

Abstract

This study compares the time-tested method obtaining people's self-reported preference measures against novel advanced neural and physiological sensing algorithms. The authors evaluate participant's liking of an "experience" product (in this case, music and movies videos) using four methods on two separate datasets: (1) the Fast Fourier Transform (FFT)-analysis of electroencephalography (EEG); (2) physiological measurements; (3) self-reported ratings; and (4) various combinations of neural and physiological signals and self-reported ratings. The DEAP and AMIGO datasets were utilized, both of which include

feelings of my honeymoon, or the experience of using a divorce lawyer's services, on a scale of one through ten? Or my love for the songs I listened to throughout those experiences, or the movies I watched that made me cry, by clicking a "like" button?



sensors



Article

M1M2: Deep-Learning-Based Real-Time Emotion Recognition from Neural Activity

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Abstract: Emotion recognition, or the ability of computers to interpret people's emotional states, is a very active research area with vast applications to improve people's lives. However, most image-based emotion recognition techniques are flawed, as humans can intentionally hide their emotions by changing facial expressions. Consequently, brain signals are being used to detect human emotions with improved accuracy, but most proposed systems demonstrate poor performance as EEG signals are difficult to classify using standard machine learning and deep learning techniques. This paper proposes two convolutional neural network (CNN) models (M1: heavily parameterized CNN model and M2: lightly parameterized CNN model) coupled with elegant feature extraction methods for effective recognition. In this study, the most popular EEG benchmark dataset, the DEAP, is utilized with two of its labels, valence, and arousal, for binary classification. We use Fast Fourier

Study Overview



- Emotion decoding pivotal for advancing **neurotechnology** and **affective computing**
 - Interconnected fields use brain data enable machines understand and respond to human emotions
- Prior electrode selection methods (often) relied on **statistical** or **spatial heuristics**
- Limits **interpretability** and **cross-dataset scalability**

Study Overview



- Comprehensive framework for EEG-based emotion recognition **reduces computational complexity**
- Efficient real-time emotion recognition in neurotechnology applications by leveraging **explainable deep learning techniques** and cross-dataset **validation**
- SHapley Additive exPlanations (SHAP) to identify most **informative electrodes** and **frequency bands** for **emotion decoding**:
 - Input parameters of prediction model reduced, reducing prediction complexity

Study Overview



- Using DEAP dataset study aimed to decode **four** core emotional dimensions: **arousal** (intensity of emotion or alertness), **valence** (pleasantness of the emotion), **dominance** (control or influence over the emotion), and **liking** (subjective preference or enjoyment).
 - **DEAP dataset:** EEG recordings from 32 participants across 32 electrodes and 8 additional physiological signals (such as electrooculogram, electromyogram, and respiration)
- For cross-dataset generalization, SEED-V and AMIGO datasets used

1. Frequency Domain Transformation for Structured Modeling

- Prepare raw EEG data for ML, signals first transformed from time domain into frequency domain using **Fast Fourier Transform (FFT)**
- Extraction of **five canonical frequency bands** (Theta, Alpha, Low Beta, High Beta, and Delta) **per electrode**, resulting in **structured tabular** format
- Transformation enabled **compatibility** w/ traditional models, preserved meaningful **spectral patterns** critical for emotion decoding

2. Modeling: Superiority of Customized Residual CNN

- Initial experiments **sequential models** like RNNs and LSTMs, limitations in capturing emotion-related temporal dynamics from raw EEG signals
- **Customized Residual Convolutional Neural Network (CNN)** developed
- Residual blocks allowed **deeper learning w/ reduced vanishing gradients**, improving feature extraction and classification
- Architecture superior performance compared to **16 popular baseline models** across four emotion dimensions

2. Modeling: Superiority of Customized Residual CNN

Model Performance Summary (Whole DEAP Dataset)

Precision	Specificity	Sensitivity	Score
0.97	0.96	0.98	0.97
Performance Gain over Baselines		32.54%	

3. Explainability with SHAP: Identifying Key Brain Regions and Bands

- Increase **interpretability** and reduce **model complexity**, SHapley Additive exPlanations (**SHAP**) used to evaluate contribution of each input feature (electrode-band pair)
- SHAP analysis revealed only subset of electrodes, particularly in the **fronto-central** and **parieto-occipital regions**, contributed significantly to **emotion recognition**
- **Delta** and **High Beta** bands play dominant roles in **encoding emotional states**

3. Explainability with SHAP: Identifying Key Brain Regions and Bands

DEAP Dataset	
Optimized Set	25 electrodes (61 features)
Performance Retention	98.78% of full model (40 electrodes 200 features)
Parameter Reduction	~24%

4. Insights from Physiological Signals

- In addition to EEG, DEAP dataset includes several **physiological signals** (e.g., eye movement, facial muscle engagement, respiration)
- SHAP revealed **non-brain signals**—particularly ocular activity, facial muscle tension (EMG), and respiration—substantial **complementary information**

4. Insights from Physiological Signals

- These signals most under Delta band, may reflect **subconscious** or **autonomic processes** linked to **emotional arousal** and **stress response**
- Combining **brain** and **peripheral physiological inputs** enhances **emotion decoding**
 - Aligns w/ **existing neuroscience literature** on brain-body interaction in affective states

5. Cross-Dataset Generalization: SEED-V and AMIGO

- Evaluate robustness of optimized feature set, mapped onto **two external datasets**: SEED-V and AMIGO
- Differences in electrode setups and target emotion labels, mapped models **retained performance** close to **full-featured counterparts**
 - **SEED-V** (6 emotions): 21 mapped electrodes (52 features) achieved 96.25% similarity to full model (64 electrodes, 320 features) while reducing parameters by 35.86%

5. Cross-Dataset Generalization: SEED-V and AMIGO

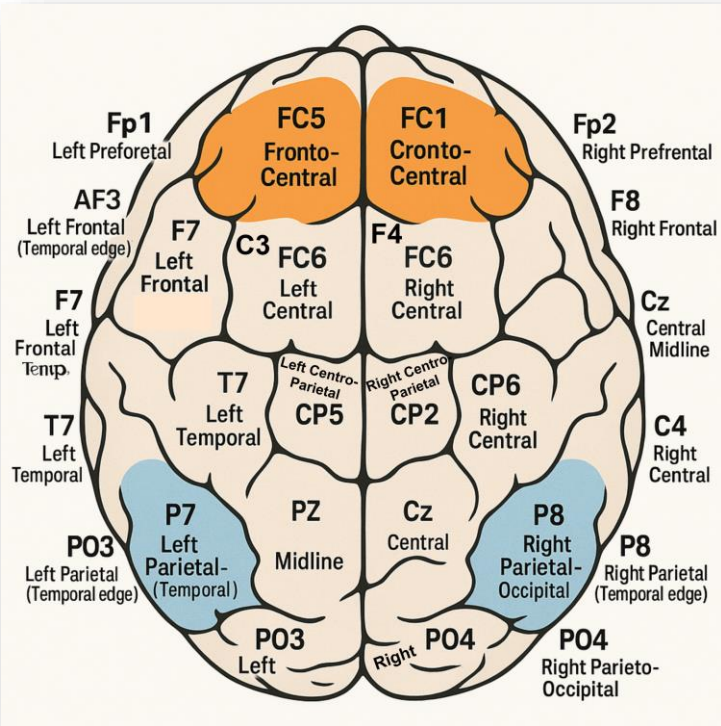
- **AMIGO** (4 emotions): 11 mapped electrodes (32 features) achieved 98.04% similarity to full model (17 electrodes, 85 features) with a 9.79% parameter reduction
- Results validate **scalability** and **transferability** of SHAP-based selection strategy, enabling efficient **emotion recognition** w/ minimal **signal overhead across datasets** and contexts

5. Cross-Dataset Generalization: SEED-V and AMIGO

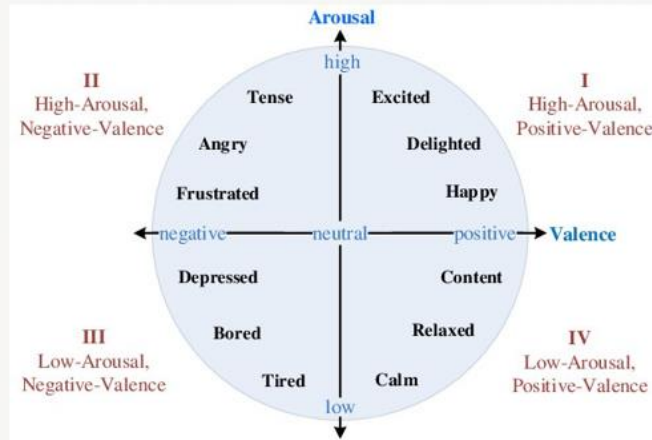
Cross-Dataset Generalization

	SEED V	AMIGO
Selected Features	21 electrodes (52 features)	11 electrodes (32 features)
Performance Retained	96.25% of full model (64 electrodes 320 features)	98.04% of full model (17 electrodes 85 features)
Parameter Reduction	~35.86%	~9.79%

Key Takeaways



Emotion Prediction:



- Fronto-central and parieto-occipital electrodes dominate emotional signal encoding, especially in the Delta and High Beta bands
- Non-brain physiological signals (eye, muscle, respiration) offer crucial affective cues and should not be overlooked, especially under the Delta band
- The proposed Residual CNN combined with SHAP-based feature pruning balances performance and interpretability
- Cross-dataset generalization (Main data: DEAP. cross data: SEED V and AMIGO) demonstrates strong potential for real-world applications in neurotechnology and affective computing, such as emotion-aware brain-computer

THANK YOU!
¡Y GRACIAS...

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